

Emphasis Interview Questions for QA Automation Role

Senior QA Interview Questions & Detailed Answers

Round 1 – Technical | Round 2 – Techno-Managerial

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3 Years Experience

Interview Date: 10 September 2026

Duration: 30 Minutes

1. Write a program to find the highest integer in an array.

CONCEPT

The objective is to find the **largest integer** present in an array. The simplest and most efficient interview approach is to traverse the array once and keep a variable that stores the current maximum value.

We initialize **max** with the first element or with **Integer.MIN_VALUE**. Then we compare every array element with max. Whenever the current element is greater than max, we update max.

This approach does not require sorting. Sorting would add unnecessary work because we only need the highest value. The one-pass solution has **O(n)** time complexity and **O(1)** extra space.

Important points to understand:

- Initialize a variable to hold the current maximum.
- Traverse every element using a for loop or enhanced for loop.
- If the current element is greater than max, update max.
- For negative values, Integer.MIN_VALUE is safer than initializing max to 0.
- A one-pass solution is preferred over sorting for this problem.
- Time complexity: **O(n)**; extra space: **O(1)**.

INTERVIEW READY ANSWER

I would find the highest integer by traversing the array only once. I initialize a variable called **max** with **Integer.MIN_VALUE**, then compare each array element with max. If the current element is greater than max, I update max. After the loop finishes, max contains the highest value. I prefer this approach because it is simple, handles negative numbers correctly, and has **O(n)** time complexity with **O(1)** extra space. I would not sort the array because sorting is unnecessary when I only need the maximum value.

IMPORTANT POINTS

1. Use a one-pass approach.
2. Initialize max correctly so negative numbers are handled.
3. Update max only when the current value is greater.
4. No sorting is required.
5. Time complexity: O(n).
6. Space complexity: O(1).

CODE / EXAMPLE

```
public class HighestInteger {
    public static void main(String[] args) {
        int[] arr = {10, 25, 7, 42, 18};
        int max = Integer.MIN_VALUE;
        for (int num : arr) {
            if (num > max) {
                max = num;
            }
        }
        System.out.println(
            "Highest integer: " + max);
    }
}
```

CODE EXPLANATION

Code / Element	Explanation
int[] arr	Stores the input integers.
int max = Integer.MIN_VALUE	Initializes max to the smallest possible int value.
for (int num : arr)	Reads each array element one by one.
if (num > max)	Checks whether the current number is greater than the current maximum.
max = num	Updates the maximum when a larger number is found.
System.out.println(...)	Prints the final highest integer.

REAL-TIME TESTING EXAMPLE

Suppose an automation suite records the execution time of several test cases in an integer array. If we need to identify the **slowest test execution**, we can use the same logic to find the highest execution-time value. The algorithm is simple, fast and does not require sorting all test results.

QUICK REVISION

Highest integer = traverse array + maintain max

For {10, 25, 7, 42, 18}, the answer is 42. Use one pass, $O(n)$ time and $O(1)$ extra space.

Testing Corner

2. Provide a detailed explanation of your framework.

CONCEPT

In a real-time automation interview, when an interviewer asks about the framework, they usually want to understand **how the project is organized, how a test starts and ends, how reusable components are designed, how test data and configuration are managed, and how execution is integrated with CI/CD and reporting.**

A typical Selenium Java framework can be designed using **Page Object Model (POM), TestNG, Maven, WebDriver**, utility classes, configuration files, test-data management, logging and reporting. The exact structure depends on the project, but the important point is to explain the responsibilities of each layer rather than simply listing tools.

The execution flow should be explained from the beginning: **configuration → driver initialization → test setup → page/object methods → assertions → reporting → cleanup.** This demonstrates that you understand the framework as an end-to-end system.

A good framework explanation also covers **reusability, maintainability, synchronization, parallel execution, failure handling, screenshots, CI execution and reporting.** These are the areas where an experienced automation engineer adds value.

Important points to understand:

- Page classes contain locators and page-specific actions.
- Test classes contain test scenarios and validations.
- Base class handles common setup and teardown.
- Utilities provide reusable operations such as waits, screenshots, Excel/JSON handling and configuration reading.
- Maven manages dependencies and build execution.
- TestNG provides annotations, assertions, grouping, data providers and parallel execution.
- Jenkins can trigger the Maven test execution as part of CI/CD.
- Reports such as Extent or Allure help the team analyze execution results.

INTERVIEW READY ANSWER

In my Selenium Java automation framework, I follow a layered and reusable approach, generally based on the **Page Object Model**. The framework is built using **Java, Selenium WebDriver, TestNG and Maven**, with supporting utilities for configuration, waits, screenshots, test data and reporting.

At the framework level, I have a **BaseTest** or driver-management layer responsible for browser initialization and common setup/teardown. I keep page-specific locators and actions inside **Page Classes**, so the test classes do not contain raw locator logic. The **Test Classes** contain business-level scenarios and assertions. Reusable operations such as explicit waits, screenshots, configuration reading and test-data handling are maintained in **utility classes**.

Maven manages the project dependencies and build lifecycle, while TestNG manages annotations, assertions, test grouping, data-driven execution and parallel execution where required. For CI/CD, Jenkins can trigger the Maven command, execute the suite and publish the generated reports. If a test fails, I capture useful evidence such as screenshots and logs. The overall goal of the framework is **reusability, maintainability, scalability and reliable execution.**

IMPORTANT POINTS

1. Use POM to separate page behavior from test scenarios.
2. Keep driver lifecycle and common setup in a base/common layer.
3. Keep test data and configuration outside hardcoded test logic.
4. Use explicit waits or reusable synchronization utilities instead of unnecessary Thread.sleep().
5. Integrate execution with Maven and CI/CD.
6. Capture screenshots/logs on failure.
7. Generate readable execution reports.
8. Design the framework so adding a new test requires minimum duplication.

CODE / EXAMPLE

```
// Typical framework structure
```

```
src/test/java
■■■ tests
■ ■■■ LoginTest.java
■■■ pages
■ ■■■ LoginPage.java
■■■ base
■ ■■■ BaseTest.java
■■■ utils
■■■ WaitUtil.java
■■■ ScreenshotUtil.java
■■■ ExcelUtil.java
■■■ ConfigReader.java

src/test/resources
■■■ config.properties
■■■ testdata.xlsx

pom.xml
testng.xml
```

FRAMEWORK CODE EXAMPLE

```
// BaseTest.java
public class BaseTest {

    protected WebDriver driver;

    @BeforeMethod
    public void setUp() {
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.get("https://example.com/login");
    }

    @AfterMethod
    public void tearDown() {
        if (driver != null) {
            driver.quit();
        }
    }

    // LoginPage.java
    public class LoginPage {

        private WebDriver driver;

        private By username =
            By.id("username");

        private By password =
            By.id("password");

        private By loginButton =
            By.id("login");

        public LoginPage(WebDriver driver) {
            this.driver = driver;
        }

        public void login(String user, String pass) {
            driver.findElement(username).sendKeys(user);
            driver.findElement(password).sendKeys(pass);
            driver.findElement(loginButton).click();
        }
    }

    // LoginTest.java
    public class LoginTest extends BaseTest {

        @Test
        public void validLogin() {
            LoginPage page =
                new LoginPage(driver);

            page.login("testuser", "password");
            // assertions
        }
    }
}
```

CODE / COMPONENT EXPLANATION

Code / Element	Explanation
BaseTest	Provides common browser setup and teardown so individual tests do not duplicate driver lifecycle code.
@BeforeMethod	Runs before each TestNG test method and prepares the test environment.
@AfterMethod	Runs after each test method and closes the browser.
LoginPage	Represents the login page and keeps its locators/actions together.
By.id(...)	Defines Selenium locators inside the page class.

Code / Element	Explanation
login(...)	Encapsulates the complete login action.
LoginTest	Contains the business-level test scenario rather than locator implementation.
Maven / pom.xml	Manages dependencies and build execution.
Jenkins	Can execute the Maven suite in a CI/CD pipeline.

REAL-TIME PROJECT EXAMPLE

For an e-commerce project, I might have **LoginPage, HomePage, ProductPage, CartPage and CheckoutPage**. A test such as **PurchaseProductTest** calls reusable page methods instead of directly writing XPath and Selenium actions in every test. If the product locator changes, I update the relevant page class instead of changing many test cases. This is one of the main benefits of POM-based framework design.

QUICK REVISION

Framework = Driver Management + POM + Tests + Utilities + Test Data + Config + Reporting + CI/CD

Explain the framework as an **end-to-end execution flow**, not merely as a list of tools. Emphasize **reusability, maintainability, scalability and failure handling**.

Testing Corner

3. Write an XPath for a given scenario.

CONCEPT

In an interview, the interviewer may give an HTML scenario and ask us to create an XPath for a particular element. The correct approach is to first identify the **stable attributes** and then construct a relative XPath that uniquely identifies the required element.

Consider a product page where several products are displayed and each product has a product name, price and an **Add to Cart** button. The interviewer asks: **Write an XPath to click Add to Cart for the product named 'iPhone 15'.**

A direct XPath based only on the button text may match multiple buttons. Instead, we can locate the product container using the product name and then navigate to its corresponding Add to Cart button. This demonstrates an understanding of **relative XPath and element relationships**.

When creating XPath, I prefer stable attributes or meaningful relationships and avoid brittle absolute paths such as `/html/body/div[2]/div[3]/...`

Important points to understand:

- First identify the unique text or attribute that identifies the required product.
- Locate the product container using that unique product name.
- Navigate from the product container to its Add to Cart button.
- Use relative XPath rather than an absolute DOM path.
- Use `contains()`, `text()`, `ancestor()`, `descendant` or following-sibling relationships when appropriate.
- Validate the XPath in browser developer tools before using it.

INTERVIEW READY ANSWER

For the given scenario, suppose the page contains multiple products and each product has its own **Add to Cart** button. If I need to click Add to Cart for **iPhone 15**, I would first locate the product container using the text **iPhone 15** and then navigate to the button within that same container. For example, if the HTML structure supports it, I can use `//div[.//h3[normalize-space()='iPhone 15']]//button[normalize-space()='Add to Cart']`.

I prefer this approach because the XPath is tied to the business identity of the product rather than the physical position of the button on the page. If the order of products changes, the XPath can still identify the correct product and its corresponding button.

IMPORTANT POINTS

1. Identify the element using stable information first.
2. Use `normalize-space()` when whitespace can vary.
3. Use a parent/ancestor container to scope the search to the correct item.
4. Avoid indexes unless there is no better stable locator.
5. Avoid absolute XPath.
6. Always verify uniqueness before using the XPath in the test.

CODE / EXAMPLE

```
// Example HTML
<div class="product-card">
<h3>iPhone 15</h3>
<span class="price">$799</span>
<button>Add to Cart</button>
</div>

<div class="product-card">
<h3>Samsung S24</h3>
<span class="price">$699</span>
<button>Add to Cart</button>
</div>

// XPath for iPhone 15 Add to Cart
//div[.//h3[normalize-space()='iPhone 15']]
//button[normalize-space()='Add to Cart']

// Selenium
driver.findElement(By.xpath(
"//div[.//h3[normalize-space()='iPhone 15']]"
+ "//button[normalize-space()='Add to Cart']"
)).click();
```

CODE / XPATH EXPLANATION

Code / Element	Explanation
<code>//div</code>	Starts by looking for div elements in the page.
<code>[.//h3[normalize-space()='iPhone 15']]</code>	Keeps only the product container that contains the required product name.
<code>normalize-space()</code>	Removes leading/trailing whitespace and normalizes text spaces.
<code>//button[normalize-space()='Add to Cart']</code>	Finds the Add to Cart button inside the selected product container.

Code / Element	Explanation
By.xpath(...)	Passes the XPath to Selenium.
click()	Clicks the identified button.
ALTERNATIVE SCENARIO	
If the HTML provides a stable product attribute such as data-product='iphone15', an even simpler XPath would be //div[@data-product='iphone15']//button[normalize-space()='Add to Cart']. In practice, I would prefer a stable attribute when the application provides one.	
REAL-TIME TESTING EXAMPLE	
In an e-commerce automation project, the same page can contain dozens of Add to Cart buttons. A generic locator such as //button[text()='Add to Cart'] may return multiple elements. Scoping the XPath to the required product prevents Selenium from clicking the wrong product's button.	
QUICK REVISION	
Business text/attribute → product container → related control For dynamic pages, prefer stable attributes and relative relationships. Avoid absolute XPath and unnecessary indexes.	

4. Write an SQL query to retrieve the details of all employees from the emp table for a given scenario.

CONCEPT

For a basic employee-data scenario, suppose the interviewer asks: **Retrieve the details of all employees from the emp table.** If the requirement is to return every column and every employee record, the query is straightforward.

The SQL keyword **SELECT** is used to retrieve data, while ***** means all columns. The **FROM emp** clause specifies the table from which the data should be retrieved.

In a real interview, I would also clarify whether 'all employees' means every row or whether a condition is required, such as employees from a particular department, employees with salary above a value, or employees hired after a specific date.

For production queries, selecting only the required columns is generally preferable to **SELECT ***, but for a question explicitly asking for all details, **SELECT *** is the direct answer.

Important points to understand:

- **SELECT** is used to retrieve records.
- The ***** symbol means all columns.
- **FROM emp** identifies the source table.
- Without a **WHERE** clause, all rows from emp are returned.
- Use **WHERE** when the interviewer adds a filtering condition.
- Use **ORDER BY** when the scenario asks for sorted output.
- For production code, select only the columns actually required when possible.

INTERVIEW READY ANSWER

If the requirement is to retrieve the details of **all employees** from the **emp** table, I would use **SELECT * FROM emp;** Here, **SELECT** retrieves the data, the asterisk represents all columns, and **FROM emp** specifies the employee table. Because there is no **WHERE** condition, the query returns every employee record in the table.

If the interviewer gives an additional condition, I would add a **WHERE** clause. For example, if they ask for employees from department 10, I would use **SELECT * FROM emp WHERE deptno = 10;** I would also mention that in real projects I prefer selecting only the required columns rather than using **SELECT *** when the requirement is specific.

IMPORTANT POINTS

1. Basic query: **SELECT * FROM emp;**
2. No **WHERE** clause means all rows are returned.
3. **WHERE** is used for filtering.
4. **ORDER BY** is used for sorting.
5. Select only required columns in production queries when possible.
6. Clarify the exact scenario if the interviewer has not provided a filtering condition.

CODE / EXAMPLE

```
-- Scenario 1: Retrieve all employee details
SELECT *
FROM emp;

-- Scenario 2: Retrieve employees
-- from department 10
SELECT *
FROM emp
WHERE deptno = 10;

-- Scenario 3: Retrieve selected details
SELECT empno, ename, job, sal
FROM emp;
```

SQL EXPLANATION

Code / Element	Explanation
SELECT *	Requests all columns from the selected table.
FROM emp	Specifies the emp table as the source.
WHERE deptno = 10	Filters the result to employees whose department number is 10.
SELECT empno, ename, job, sal	Retrieves only the specified employee columns.
;	Marks the end of the SQL statement.

REAL-TIME TESTING EXAMPLE

7. Explain method overloading and method overriding with examples.

CONCEPT

Method overloading and **method overriding** are both related to polymorphism, but they solve different problems. Overloading means defining multiple methods with the **same name but different parameter lists** in the same class or related context.

Overriding means a child class provides a new implementation of a method inherited from its parent class.

Overloading is resolved at **compile time**, so it is commonly called **compile-time polymorphism**. The compiler selects the method based on the number, type or order of parameters.

Overriding is resolved at **runtime**, so it is commonly called **runtime polymorphism**. The method that executes depends on the actual object created, even when the reference is of the parent type.

A key interview point is that changing only the **return type** is not sufficient for overloading. For overriding, the child method must follow the parent method's compatible signature and cannot reduce access visibility.

Important points to understand:

- **Overloading**: same method name + different parameter list.
- **Overriding**: child class redefines inherited method with the same signature.
- Overloading → compile-time polymorphism.
- Overriding → runtime polymorphism.
- Overloading can differ by number, type or order of parameters.
- Return type alone cannot create an overloaded method.
- Use `@Override` to clearly indicate an overridden method.

INTERVIEW READY ANSWER

Method overloading means having multiple methods with the **same name but different parameter lists**. The difference can be in the number, type or order of parameters. It is resolved by the compiler, so it is called **compile-time polymorphism**.

Method overriding occurs when a **child class provides its own implementation** of a method inherited from the parent class using the same method signature. The decision about which implementation to execute is made at runtime based on the actual object, so it is called **runtime polymorphism**.

For example, in an automation utility I could overload a click method so it accepts either a By locator or a String locator. For overriding, a specialized browser/page class could provide its own implementation of a common action defined in a parent class. I normally use `@Override` because it makes the intent clear and lets the compiler validate the method signature.

IMPORTANT POINTS

1. Overloading changes the parameter list; overriding keeps the compatible method signature.
2. Overloading is compile-time; overriding is runtime.
3. Return type alone cannot overload a method.
4. A child class uses `@Override` for an overridden method.
5. Overriding supports runtime polymorphism.
6. Both concepts improve flexibility and reusable object-oriented design.

CODE

```
// Method Overloading
class SeleniumUtility {

    public void click(By locator) {
        System.out.println(
            "Click using By locator");
    }

    public void click(String locator) {
        System.out.println(
            "Click using String locator");
    }

    public void click(By locator,
        int timeout) {
        System.out.println(
            "Click with timeout: " + timeout);
    }
}
```

```
// Method Overriding
class BasePage {

    public void openPage() {
        System.out.println(
            "Opening base page");
    }
}

class LoginPage extends BasePage {

    @Override
    public void openPage() {
        System.out.println(
            "Opening login page");
    }
}

public class TestRunner {

    public static void main(String[] args) {

        BasePage page =
            new LoginPage();

        page.openPage();
    }
}
```

CODE EXPLANATION

Code / Element	Explanation
click(By locator)	Overloaded method accepting a By locator.
click(String locator)	Same method name but a different parameter type, so it is an overload.
click(By locator, int timeout)	Same method name with an additional parameter, another overload.
BasePage.openPage()	Parent class defines the common behavior.
LoginPage extends BasePage	Child class inherits the parent method and can replace its implementation.
@Override	Explicitly marks the child implementation as an override.
BasePage page = new LoginPage()	Parent reference points to a child object, enabling runtime polymorphism.
page.openPage()	Runtime dispatch invokes LoginPage's overridden method.

REAL-TIME AUTOMATION EXAMPLE

For **overloading**, a framework utility might provide **click(By locator)**, **click(String locator)** and **click(By locator, int timeout)**. The test can call the same method name with different inputs. For **overriding**, a common BasePage method such as openPage() can be specialized by LoginPage, ProductPage or CheckoutPage when each page requires different behavior.

COMPARISON

Point	Overloading	Overriding
Meaning	Same method name, different parameters	Child provides a new implementation
Polymorphism	Compile-time	Runtime
Parameters	Must be different	Compatible/same signature
Inheritance required	No	Yes, normally parent-child relationship
Return type	Return type alone cannot overload	Must be compatible with parent method
Annotation	No special annotation	@Override recommended

QUICK REVISION

Overloading → same name + different parameters → compile time.

Overriding → child redefines parent method → runtime.

Automation example: **click(By)** and **click(String)** demonstrate overloading; **LoginPage.openPage()** replacing **BasePage.openPage()** demonstrates overriding.

Thank You

Mphasis Interview Preparation

QA Automation Role